

Vikram Chennamsetty

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Professional Summary

B.Tech Electronics & Communication Engineering student specializing in software engineering across backend systems (Python, FastAPI), full-stack web development (Next.js, TypeScript, Supabase/PostgreSQL), and AI/LLM integration (Ollama, vector memory). Built a production full-stack ordering platform with live payment verification and a validated local-first AI software platform. Seeking backend, full-stack, or AI/LLM engineering roles.

Technical Skills

- **Programming:** Python, C, JavaScript/TypeScript, SQL
- **Backend & APIs:** FastAPI, REST API design/integration, Authentication
- **Frontend / Full-Stack:** Next.js, React, HTML, CSS
- **Databases:** PostgreSQL, Supabase, SQLite
- **AI / LLM:** Ollama (local LLM execution), LLM integration, vector memory (ChromaDB), AI-assisted automation
- **Developer Tools:** Git, GitHub, VS Code
- **Cloud / Deployment:** Vercel
- **IoT / Embedded:** ESP32, Arduino, LoRa, Embedded C, Sensors

Selected Projects

JARVIS – AI Software Engineering / Local-First AI Assistant Platform *Python, FastAPI, Ollama, SQLite, ChromaDB*

- Architected a local-first AI assistant with a FastAPI backend, multi-agent/tool workflows, and an MCP-based tool registry governed by a permission/policy layer.
- Integrated local LLM execution via Ollama with ChromaDB vector memory (SQLite-backed) for persistent context, plus a voice pipeline (faster-whisper/VAD for speech-to-text, Piper/Kokoro/pyttsx3 for text-to-speech) and Electron desktop & Flutter Android clients over the REST API backend.
- Completed Phase 1 validation – 33 automated tests passing, backend issues resolved, and acceptance criteria met ahead of Phase 2.

Ammu's Chai With Maska Bun – Security-Hardened Full-Stack Food Ordering Platform *Next.js 15, TypeScript, Supabase, PostgreSQL, Razorpay*

- Built and deployed a production ordering platform ([live demo](#)) with Next.js 15, TypeScript, React, and Supabase/PostgreSQL using Row-Level Security and PL/pgSQL triggers.
- Implemented Razorpay checkout with server-side HMAC-SHA256 signature verification, routing privileged transaction/order writes through server-side actions rather than the client; live checkout, signature verification, and order creation confirmed on a real successful production payment. Webhook handling is implemented but not yet live-tested.
- Added authentication, a persistent cart (Zustand), Zod-validated forms (React Hook Form), an admin dashboard, and Resend/Telegram notifications; shipped as an installable PWA (Serwist) on Vercel.

AI-Driven Predictive Cooling for Data Centers *Python, AI/ML, Digital Twin, Predictive Control, Data Processing*

- Built a physics-based digital-twin simulation of a 40-rack data center (thermal recirculation model, CPU/GPU workload generator) with a bidirectional LSTM forecasting peak rack temperature at 5/10/15-min horizons and Monte Carlo-dropout uncertainty estimation.
- Implemented an uncertainty-aware MPC-style controller (dynamic, uncertainty-scaled safety margin) exposed via FastAPI, benchmarked against Static and Reactive baselines over a 3-day/48-hour simulated evaluation.
- Cut peak rack temperature to 22.84°C (vs. 25.66°C static) with zero safety violations, trading off higher cooling energy for that margin – a documented safety/efficiency trade-off (simulation-only).

Elephant Intrusion Detection and Crop Protection System *ESP32-CAM, Arduino UNO, LoRa (SX1278), Embedded C*

- Designed a perimeter intrusion detection system using PIR/ultrasonic sensors on Arduino UNO with an ESP32-CAM module for motion-triggered image capture, integrating sensor, relay, and wireless modules with a supporting PCB schematic in KiCad.
- Prototyped LoRa-based (SX1278) long-range alert transmission for remote-area intrusion alerts; core logic simulated and partially tested.

Education

B.Tech, Electronics and Communication Engineering

2023 – 2027

Specialization: Signal Processing | CGPA: 8.43

Kalasalingam Academy of Research and Education

Experience

Emertxe Information Technologies – IoT Intern

- Completed an industry-oriented IoT internship covering end-to-end IoT architecture, sensor/actuator integration, embedded programming, device communication, cloud telemetry, and SDLC-based product development.
- Trained in Embedded C/C++, Python automation, Arduino-based simulation, IoT networking, cloud telemetry, and dashboard-based monitoring through practical Industry 4.0 workflows, with the capstone focus on Edge AI Based Smart EV Charging Station Optimizer.

Achievements & Leadership

- Google Gemini Student Ambassador – selected to support AI awareness activities on campus.
- Advanced to Round 3 in multiple campus IoT innovation hackathons with A-grade technical evaluations.